4TH OPEN SCIENTIFIC INTELLIGENCE (OSI) HACKATHON 2026 · OFFICIAL 1-PAGE PROJECT WRITE-UP

PolyVAN-Agent: Repurposing Autoregressive Attention as a Variational Partition Function Solver for Polyelectrolyte–Peptide Assembly & Formulation Design

Yansong Li · South China University of Technology (SCUT) · Code, Data & Interactive Web Workbench: PolyVAN-OSI-Hackathon-2026

Abstract — In functional soft-matter formulations—from biomedical lipopeptide–hyaluronan (HA) carriers (3 ppm Pal-KTTKS) to industrial papermaking wet-end retention aids (0.5–5 ppm cationic PAM)—*total additive dosage systematically diverges from active species concentration*. At micromolar operating concentrations, all-atom molecular dynamics faces a severe "concentration chasm" (>700 nm simulation boxes), while classical 1D nearest-neighbor transfer matrices (L0 DP) fail when semi-flexible chains form spatial loops that couple sequence-distant hydrophobic ligands. Here we present **PolyVAN-Agent**, an open scientific intelligence framework that (1) exploits the exact **mathematical isomorphism between GPT causal autoregressive decomposition and statistical-mechanical Boltzmann distributions** to solve loop-coupled polyelectrolyte partition functions without training data, and (2) embeds this solver as a verified tool inside an LLM experimental-design agent. Against exact 2^{20} (1,048,576-state) enumeration, our causal Transformer achieves **0.00% free-energy error (0.0001 k_BT)** and importance-sampling **ESS = 1.00**, proving that ignoring long-range loops underestimates site occupancy by **13.8%**. Across a 200-sample prior ensemble, PolyVAN-Agent discovers a sharp **oligosaccharide geometric cliff at dp20 (4.0 kDa)** and a **150 mM salt-induced cluster reset**, enabling automated design of falsifiable ITC/fluorescence protocols (E1–E5) and salt-switchable formulations.

1. PHYSICAL BOTTLENECK: LOOPS BREAK TRANSFER MATRICES

Polyelectrolyte–amphiphile formulations operate at trace dosages (3 ppm ≈ 3.74 µM Pal-KTTKS on 0.1 wt% HA), where functional efficacy depends on equilibrium partitioning among four states: *free monomer*, *solution self-aggregate*, *isolated chain-bound monomer*, and *chain-templated multivalent clusters*.

While 1D dynamic programming (L0 transfer matrix) captures nearest-neighbor cooperativity (ω) and finite chain length (N), HA is a semi-flexible worm-like chain (persistence length $l_p \approx 5$ nm). Sequence-distant sites $|i-j| \geq 3$ contact via spatial looping with probability $P_{\text{loop}}(|i-j|) \propto |i-j|^{-3/2}$. Because C16 palmitoyl tails carry strong hydrophobic association ($\Delta G \approx -15$ to -25 kJ/mol, Boltzmann factor 4.2×10^2–2.4×10^4), non-local loop couplings dominate the partition function Z, causing 1D transfer matrices to break down.

2. ISOMORPHISM: GPT ATTENTION AS PHYSICS SOLVER

A lattice polymer with site occupancies $\mathbf{s} = (s_1, \ldots, s_N) \in \{0, 1\}^N$ obeys the exact autoregressive probability decomposition:

$$q_\theta(\mathbf{s}) = \prod_{i=1..N} q_\theta(s_i \mid s_1, \ldots, s_{i-1}) \iff \textbf{Causal GPT Token Generation}$$

Rather than fitting external datasets, we train a 2-layer causal Transformer (`MiniGPT`, $d = 48$, $h = 4$) directly on the loop-coupled Hamiltonian $E(\mathbf{s}) = -\mu\sum s_i - J\sum s_i s_{i+1} - J_{\text{long}}\sum_{j-i\geq3}|i-j|^{-3/2} s_i s_j$ by minimizing the variational free energy upper bound via β-annealed REINFORCE:

$$F_q = \mathbb{E}_{\mathbf{s}\sim q_\theta}[\, E(\mathbf{s}) + k_B T \ln q_\theta(\mathbf{s}) \,] \geq F_{\text{exact}} = -k_B T \ln Z$$

Causal self-attention naturally captures all-to-all long-range site couplings while supporting exact $\mathcal{O}(N)$ independent microstate sampling and unbiased importance-sampling reweighting (F_{IS}).

3. EXACT 2^{20} BENCHMARK & QUANTITATIVE PROOF

Solver / Regime (N = 20, 1.05×10^6 states)	F_{exact} (k_BT)	Solver F	Rel. Err	Occupancy	ESS
Nearest-Neighbor (J_{long} = 0) · MiniGPT	−20.0460	**−20.0460**	**0.00%**	0.8736 / 0.8728	1.00
+ Loop Coupling (J_{long} = 3) · MiniGPT	−49.4726	**−49.4725**	**0.00%**	0.9944 / 0.9943	1.00
+ Loop Coupling · Classical L0 DP	−49.4726	**−20.0460**	**59.5%**	**0.8736 (−13.8%)**	—

`MiniGPT` matches exact 2^{20} enumeration to 10^{-4} k_BT with ESS = 1.00, whereas classical L0 DP misses 29.43 k_BT of loop stabilization and underestimates site occupancy by **13.8%**.

4. PHYSICAL DISCOVERY: DP20 CLIFF & SALT RESET

Coupling our partition solver with a 200-sample prior sensitivity ensemble over binding stoichiometry ($n_{\text{site}} \in \{1, 2, 3\}$ disaccharides/peptide) and minimum nucleus size ($n_{\text{nuc}} \in \{3..8\}$) reveals two falsifiable physical rules:

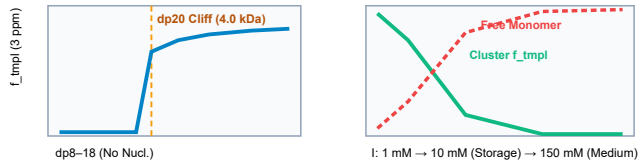


Fig 1. (Left) Sharp geometric step at dp = $2\cdot n_{\text{site}}\cdot n_{\text{nuc}}$ = 20 (4.01 kDa): effective CAC drops from >285 ppm (dp18) to 2.32 ppm (dp20). (Right) Counterion-release salt switch (I = 1→150 mM): 73.5% templated clusters at 10 mM dissociate into >99% free monomer at 150 mM.

- **Geometric Oligosaccharide Cliff (E2):** Across 168/200 templating priors, $P(\text{MW}_{90} < 20\text{ kDa}) = 97.6\%$ (median 6.4 kDa). Chains shorter than $n_{\text{site}} \times n_{\text{nuc}}$ disaccharides cannot host a nucleus, creating a 100-fold CAC drop between dp18 and dp20.
- **Culture-Medium Reset & Salt-Switchable Carrier (E3/E4):** Because binding scales as $K(I) \propto I^{-Z_{\text{rel}}}$, diluting a low-salt formulation (I = 10 mM, f_{tmpl} = 73.5%) into physiological medium (I = 150 mM) melts >90% of clusters in 76.2% of priors—enabling a patentable **"Storage-Locked, Skin-Released" (A+D) formulation**.

5. AGENTIC ORCHESTRATION & CROSS-DOMAIN IMPACT

PolyVAN-Agent wraps the Python/JS partition solvers and 200-prior sensitivity engine into an interactive Web Workbench covering two isomorphic industrial domains:

Isomorphic Feature	Domain A: HA + Pal-KTTKS	Domain B: Papermaking CPAM
Operating Dosage	3.0 ppm lipopeptide	0.5–5.0 ppm (100–500 g/t pulp)
Active Species	Templated cluster / monomer	Loop-and-tail bridging state
Parasitic Sink	β-sheet fibril precipitation	Anionic trash (DCS) complex
Master Variable	Ionic strength I (1–150 mM)	White-water conductivity κ
Ground-Truth Assay	ITC (E1) + Pyrene CAC (E2–E4)	Filtrate PCD + DDJ (TAPPI T261)

References: [1] Wu, Wang & Zhang, *Phys. Rev. Lett.* **122**, 080602 (2019). [2] Hubbe et al., *BioResources* **4**, 405 (2009). [3] Castelletto et al., *Soft Matter* (Pal-KTTKS self-assembly).